

Early-Exit Graph Neural Networks for Link Prediction

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TL;DR

GNNs are effective for link prediction, but they often use the same amount of computation for every node/link. This is inefficient. Our approach allows nodes to stop updating once further message passing is unlikely to help. We compare independent node-level exiting with a more stable neighbourhood-aware variant.

Main takeaway: early exiting can reduce theoretical GNN compute for link prediction without consistently hurting quality, but stable training remains the main challenge.

Prior Work

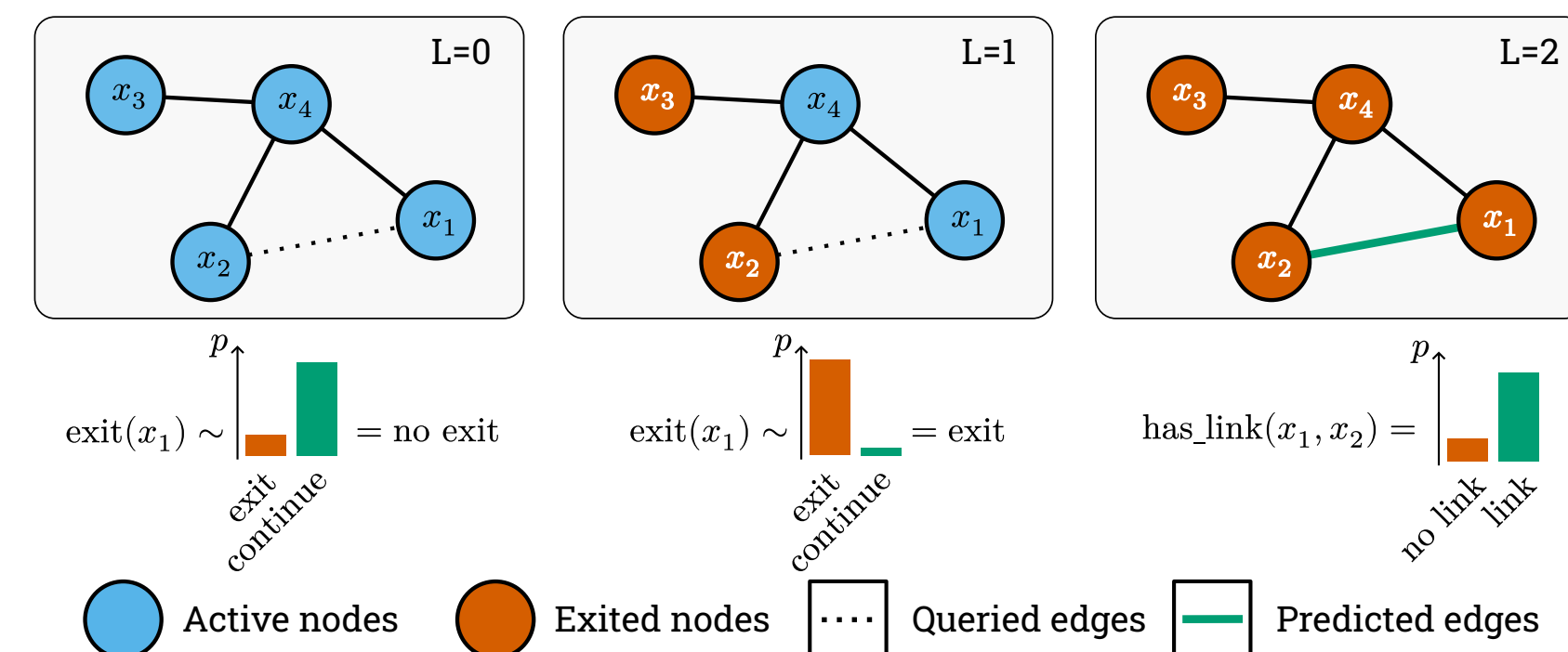
We base our work on the prior early-exit work by Di Francesco [1], and the evaluation is done via HeaRT framework [2]. To our knowledge, we are the first to work on early-exiting for link prediction.

The choice of Gumbel estimator was originally borrowed from [1].

[1] A. G. Di Francesco et al., "Early-Exit Graph Neural Networks," arXiv:2505.18088, 2025.

[2] J. Li et al., "Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking," NeurIPS, vol. 36, pp. 3853–3866, 2023.

Node Early Exiting

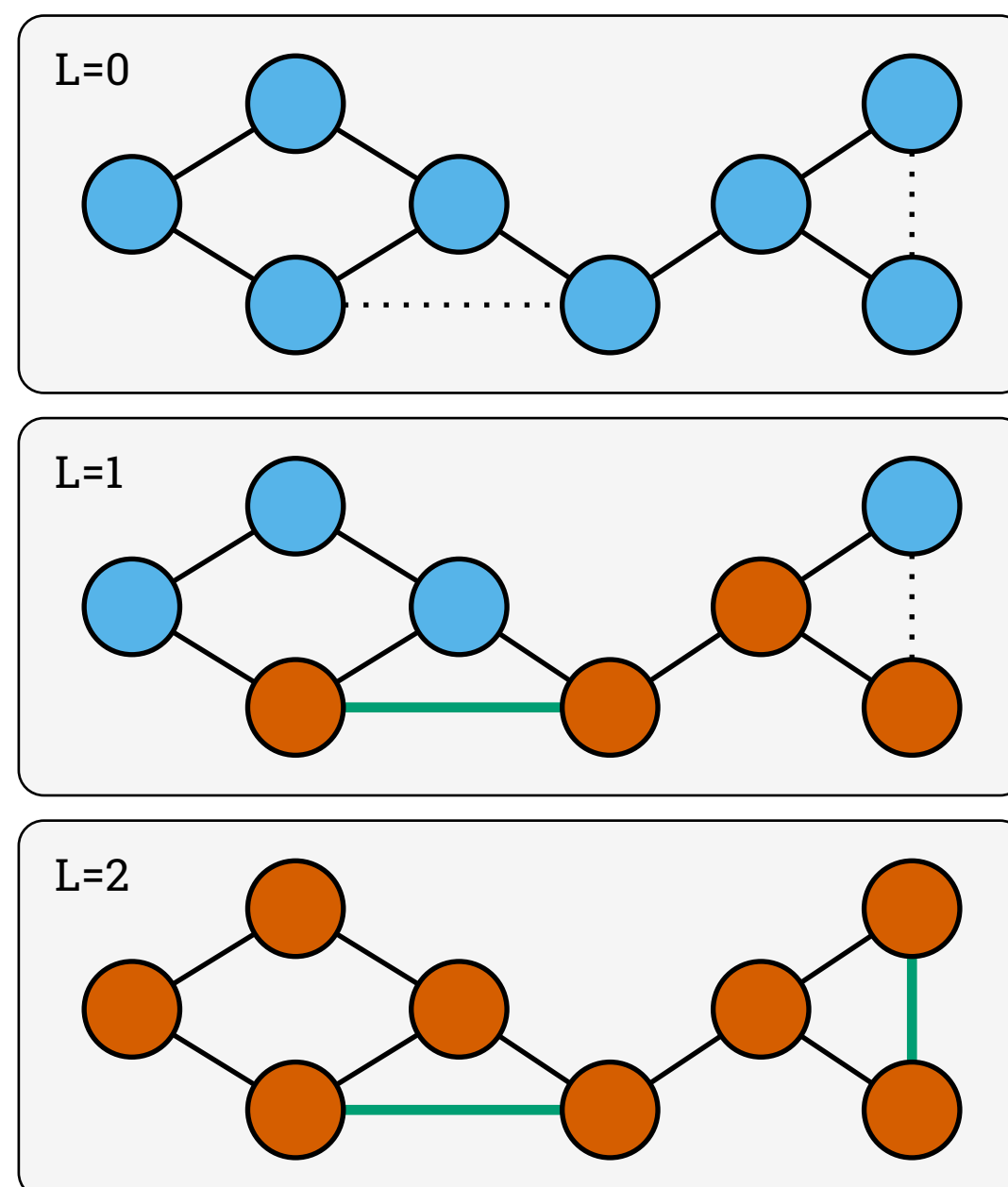


Attach a special early-exiting head to each node; train with a straight-through gradient estimator.

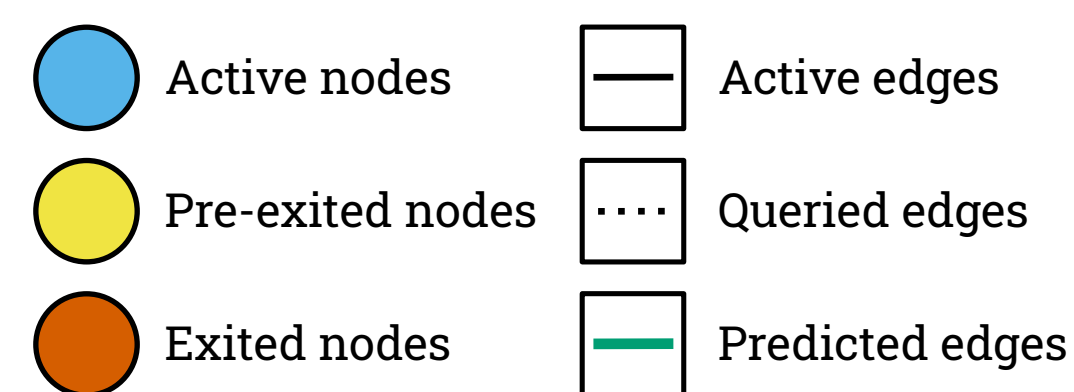
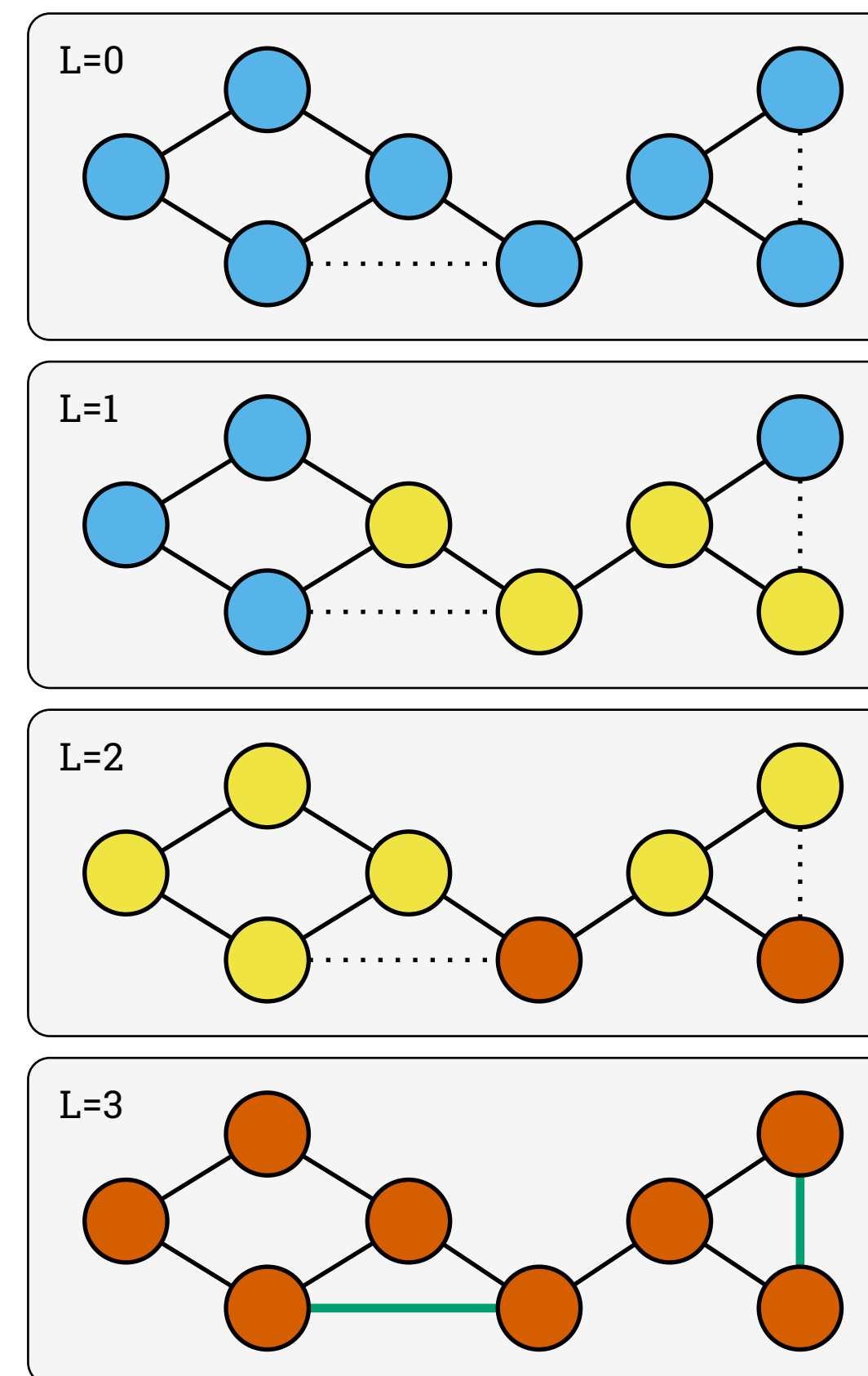
$$\mathbf{c}_u^l = \text{Softmax}\left(\frac{\log \mathbf{C}_u^l + \mathbf{g}_u^l}{\nu_u^l}\right), \quad \mathbf{g}_u^l \sim \text{Gumbel}(0, 1),$$

Subgraph Early Exiting

Node-based exiting

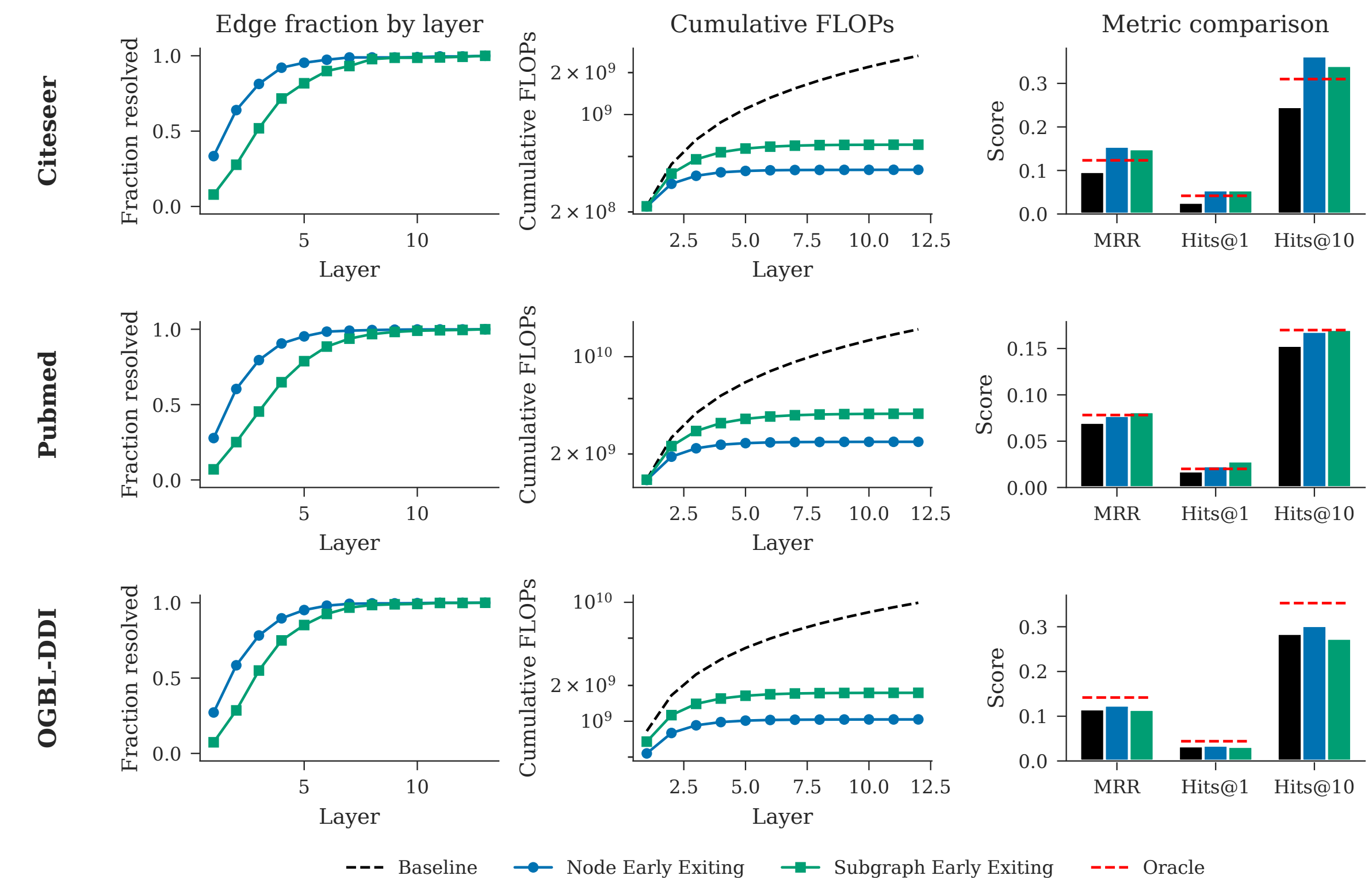


Subgraph-based exiting



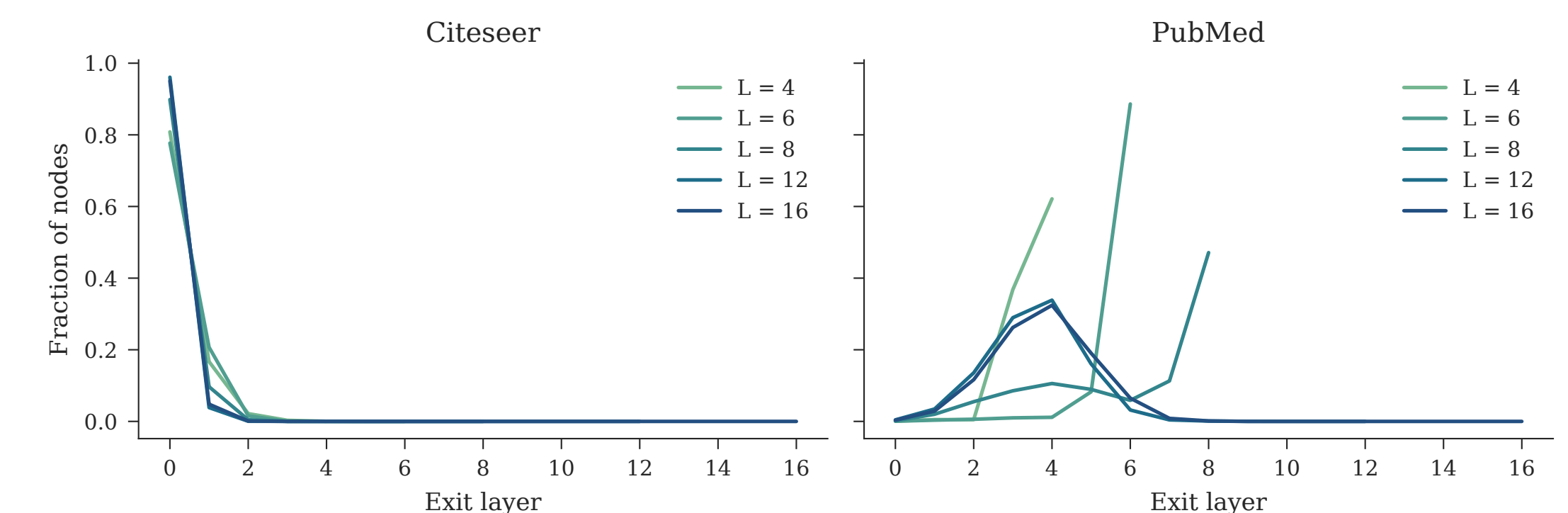
Have two per-layer heads; one is "done" indicator, another is "exit" indicator. At each depth, the "exit" head takes in an aggregate of "done" heads outputs, and predicts whether to exit.

Results



There is no collapse in the exiting layer: average exit depth is 2-3 for most of the problems. The savings in theoretical FLOPs are around 4x vs 12-deep GCN. Our methods outperform oracle-depth GCN on Citeseer, match on PubMed and underperform on OGBL-DDI.

Limitations



The main limitation is the instability of the approach. Lacking an explicit depth-disencouraging loss there is no stabilization, and thus one can alter the behaviour of the model by simply tweaking various hyperparameters. For instance, one can cause the model to collapse to either never-exiting or instantly-exiting by changing the learning rate. This limits practical applications, and the main issue we intend to address in future work.